Do Now · retrieval

Lesson 053 · Graphs: Adjacency Lists and Traversal

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In a BST, where are keys smaller than the parent?

Q2

What is a graph vertex?

Q3

How does a queue differ from a stack?

Q4

An undirected graph has vertices A, B, C, D and edges A–B, A–C, C–D. In vertex order A,B,C,D, write row C of its adjacency matrix and explain why the matrix is symmetric.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Discover — An adjacency list records each vertex and its neighbours. For undirected graphs, an edge is represented in both vertices' neighbour lists.

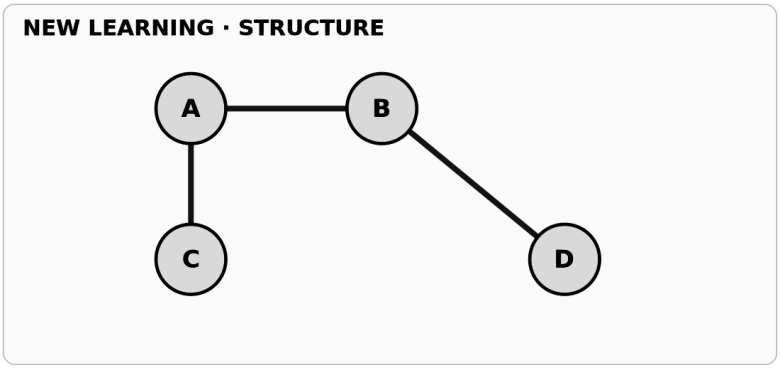
Core idea 2

Trace — For BFS, enqueue a starting vertex, mark it visited, and repeatedly dequeue and enqueue undiscovered neighbours.

Core idea 3

Evaluate — Traversal order depends on neighbour ordering. A visited set prevents infinite revisiting when the graph contains cycles.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

graph — Vertices linked by edges.

adjacency list — Neighbours indexed by vertex.

BFS — Breadth-first search using queue.

visited — Vertices already discovered.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

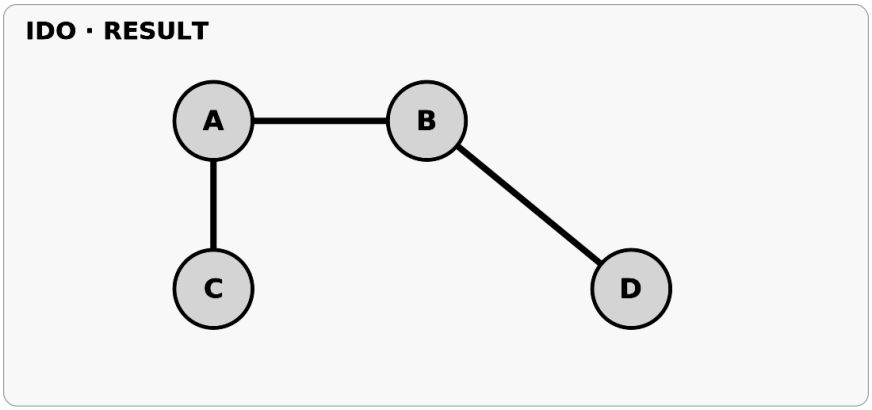
Problem

Trace BFS from A given neighbours A:[B,C], B:[A,D], C:[A], D:[B].

Follow the worked method

1. Explain the trace — With B before C, BFS visit order A,B,C,D. Each vertex is marked on enqueue.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the trace
BFS visit order X,Y,Z,W. FIFO queue processes all distance-one vertices before W.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Write adjacency lists for undirected edges A-B,A-C,B-D. [6]

Question 2

Trace BFS from A in the lesson graph, recording the queue after expanding A and B. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why DFS can use a stack while BFS uses a queue and why cycles require visited tracking. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In an undirected graph, A–B is normally stored only under A in a basic adjacency list.

Use key vocabulary from memory

1. Explain graph in your own words.

2. Explain adjacency list in your own words.

3. Explain BFS in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.